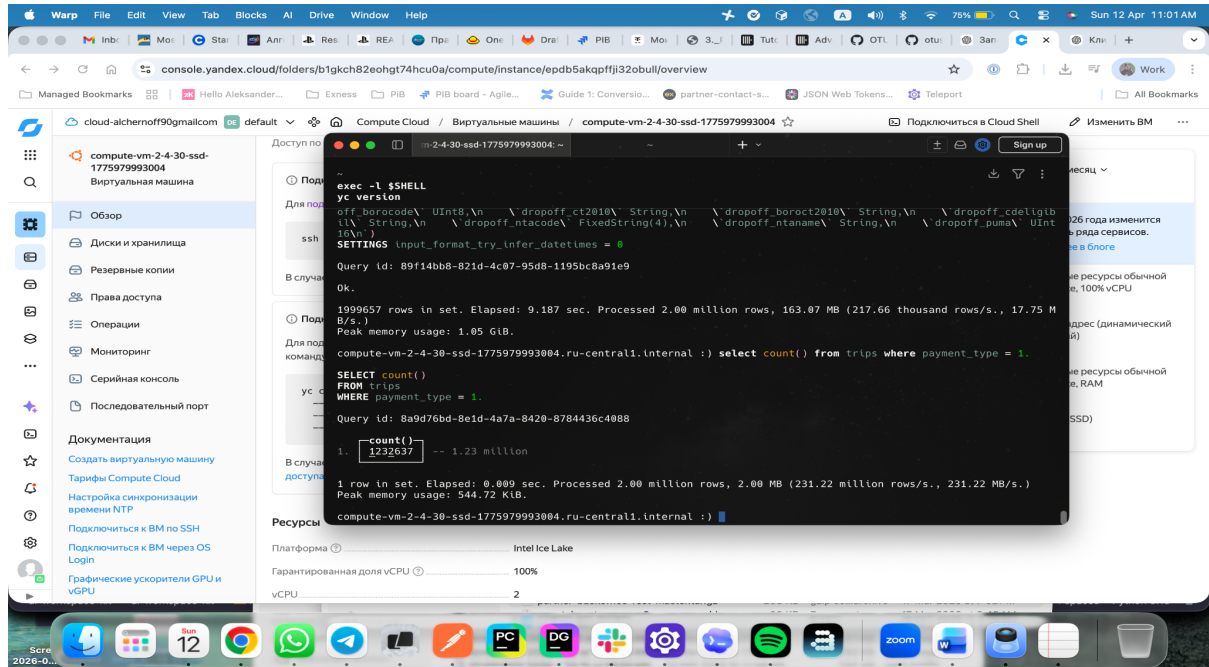


ClickHouse Performance Lab Report

1. Installation and Setup

ClickHouse was installed on a Yandex Cloud VM (2 vCPU, 4GB RAM, 30GB SSD).



```
compute-vm-2-4-30-ssd-1775979993004.ru-central1.internal :) select count() from trips where payment_type = 1.  
  
SELECT count()  
FROM trips  
WHERE payment_type = 1.  
  
Query id: 8a9d76bd-8e1d-4a7a-8420-8784436c4088  
  
1. count() -- 1.23 million  
   1232637  
  
1 row in set. Elapsed: 0.009 sec. Processed 2.00 million rows, 2.00 MB (231.22 million rows/s., 231.22 MB/s.)  
Peak memory usage: 544.72 KiB.  
  
compute-vm-2-4-30-ssd-1775979993004.ru-central1.internal :)
```

Обзор

Идентификатор..... epdb5akqpffji32obull

Статус..... Running

Имя..... compute-vm-2-4-30-ssd-1775979993004

Дата создания..... 12.04.2026, в 10:47

Внутренний FQDN..... compute-vm-2-4-30-ssd-1775979993004.ru-central1.internal 

Зона доступности ..... ru-central1-b

```
. (AUTHENTICATION_FAILED) (version 26.3.7.8 (official build))
yc-admin@compute-vm-2-4-30-ssd-1775979993004:~$ echo "SELECT count() FROM trips WHERE payment_type = 1" | \      clickhouse-benchmark \
--host localhost \
--port 9000 \
--user default \
--password password \
-i 10 -c 4
Loaded 1 queries.

Queries executed: 10 (100%).

localhost:9000, queries: 10, QPS: 50.890, RPS: 101763194.027, MiB/s: 97.049, result RPS: 50.890, result MiB/s: 0.000.

0%          0.008 sec.
10%         0.014 sec.
20%         0.018 sec.
30%         0.021 sec.
40%         0.022 sec.
50%         0.024 sec.
60%         0.024 sec.
70%         0.024 sec.
80%         0.025 sec.
90%         0.025 sec.
95%         0.028 sec.
99%         0.028 sec.
99.9%       0.028 sec.
99.99%      0.028 sec.
```

```

yc-admin@compute-vm-2-4-30-ssd-1775979993004:~$ echo "SELECT count() FROM trips WHERE payment_type = 1 SETTINGS max_threads = 1,
max_block_size = 8192, use_uncompressed_cache = 0" | clickhouse-benchmark --host localhost --port 9000 --user default -
-password password -i 10 -c 4
Loaded 1 queries.

Queries executed: 10 (100%).

localhost:9000, queries: 10, QPS: 50.121, RPS: 100224444.029, MiB/s: 95.581, result RPS: 50.121, result MiB/s: 0.000.

0%          0.014 sec.
10%         0.016 sec.
20%         0.021 sec.
30%         0.023 sec.
40%         0.026 sec.
50%         0.026 sec.
60%         0.026 sec.
70%         0.028 sec.
80%         0.028 sec.
90%         0.030 sec.
95%         0.035 sec.
99%         0.035 sec.
99.9%       0.035 sec.
99.99%      0.035 sec.

```

2. Query Execution

Query: `SELECT count() FROM trips WHERE payment_type = 1`

3. Benchmark

Benchmark executed using `clickhouse-benchmark` with 10 iterations and 4 concurrent connections.

4. Optimization Test

Parameters changed:

- `max_threads = 1`
- `max_block_size = 8192`
- `use_uncompressed_cache = 0`

Conclusion: performance slightly degraded due to reduced parallelism and disabled cache.